

PAPER_2122 — Triple $\{\beta_i, \rho_{\text{vac}}, c\}$: First Projection-Layer 3-Constant Instance (REVISED)

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REVISED IN PLACE 2026-07-22 — projection-layer reclassification per canonical F_UBi/F_UBii/k_spring/quantum-chain deepsearch. Convergence statements scoped to the QCalc projection layer (wrap classes annotated 'framework wrap only'). Canonical layer documented below was fully primitive prior to and independent of these events.

Result

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Ubi = beta_i * rho_vac * V * c**2 * SCm (wrap, energy form)
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First projection-layer triple. β_i canonicalized $0.6 \rightarrow 0.6029$ (Rule 2). PAPER_2121 forecast fulfilled 15 rounds early.

Key Reinterpretation — the Projection Identity

$\{\rho_{\text{vac}}, c\}$ is the projection pair: $\rho_{\text{vac}} \cdot c^2$ is exactly the canonical quantum chain's $\rho_m \rightarrow \rho_E$ conversion ($\rho_E = \rho_m \cdot c^2$, line 33295). Canonical buoyancy $F_{\text{UBi}} = (\rho_{\text{SCm}} \cdot M/r) \cdot (1 + \beta_i \cos + \text{SSq})$ contains NO c , NO $\rho_{\text{vac}} \cdot V \cdot c^2$. β_i is the **cross-layer invariant** — the same locked 0.6029 in canonical kernel $\{\rho_{\text{SCm}}, \beta_i\}$ and in the wrap. ω_{SCm} and Φ_{res} appear in ZERO wraps — they live in k_{spring} , where canonical buoyancy gets its scale.

The Canonical Layer

The canonical layer (uqff_pure_calculator.py, wired): $F_{\text{UBi}} = (\rho_{\text{SCm}} \cdot M/r) \cdot (1 + \beta_i \cos(\pi t_n) + \text{SSq})$; PAPER_1203 pair with $k_{\text{spring}} = (\rho_{\text{UA}}/\rho_{\text{SCm}}) \cdot \omega_{\text{SCm}} \cdot \Phi_{\text{res}} = 1.05\text{e}13$; dynamic $\beta(t, E, Z)$; 26-level quantum chain with $\rho_E = \rho_m \cdot c^2$ (line 33295) and 633333.333 validation; r_{hz} equilibrium residual $< 1\text{e}-10$. Canonical buoyancy kernel: $\{\rho_{\text{SCm}}, \beta_i\}$. The wrap's $\rho_{\text{vac}} \cdot c^2$ is the chain's mass-to-energy conversion — $\{\rho_{\text{vac}}, c\}$ is the PROJECTION PAIR, not a physics kernel.